

WHICH REAL QUADRATIC PLANES NEED FOUR COLOURS

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ABSTRACT. For a subfield $F \subset \mathbb{R}$ let $\chi(F^2)$ be the chromatic number of the graph on F^2 that joins two points at Euclidean distance 1. For a real quadratic field $\mathbb{Q}(\sqrt{d})$, d squarefree, it is known that $\chi(\mathbb{Q}(\sqrt{d})^2) = 2$ unless $d \equiv 3 \pmod{4}$, that $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 3$ if $d \equiv 3 \pmod{4}$, and that $\chi(\mathbb{Q}(\sqrt{d})^2) \leq 3$ unless $d \equiv 11 \pmod{12}$. We prove that $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for every $d \equiv 11 \pmod{12}$. So the plane over $\mathbb{Q}(\sqrt{d})$ needs four colours exactly when $d \equiv 11 \pmod{12}$, and, with the upper bounds of Fischer and Moorhouse, $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for all these d except possibly $d \equiv 47, 143, 167 \pmod{168}$. The proof exhibits, for each d , a finite set of unit vectors, made of one or two vectors and their rotations by the rational rotations whose denominators divide a power of 5, and shows that no character of the group they generate maps all of them into $[1/3, 2/3]$; a characterisation of 3-colourable Cayley graphs of abelian groups by such characters then excludes 3-colourings. The main step is an exact description of the characters of $\frac{1}{N}\mathbb{Z}[i]$, $N = 5^k$, that keep every rational rotation with denominator dividing N in $[1/3, 2/3]$.

1. INTRODUCTION

Let $F \subset \mathbb{R}$ be a field. The *plane over F* is the graph on F^2 in which x and y are adjacent when $|x - y| = 1$, and $\chi(F^2)$ is its chromatic number. For $F = \mathbb{R}$ this is the Hadwiger–Nelson problem, and $5 \leq \chi(\mathbb{R}^2) \leq 7$ [1]. For $F = \mathbb{Q}$, $\chi(\mathbb{Q}^2) = 2$ [10]. For real quadratic fields $K = \mathbb{Q}(\sqrt{d})$, $d \geq 2$ squarefree, the following was known:

- $\chi(K^2) = 2$ if $d \not\equiv 3 \pmod{4}$ [3, 2, 5];
- $\chi(K^2) \geq 3$ if $d \equiv 3 \pmod{4}$ [2, Thm. 8] (and [5, Thm. 8.6] when K contains $\sqrt{p}$ for a prime $p \equiv 3 \pmod{4}$); for $u = ((1 - d)/(1 + d), 2\sqrt{d}/(1 + d))$ and its mirror image $\bar{u}$, the relation $\frac{d+1}{4}(u + \bar{u}) + \frac{d-1}{2}(1, 0) = 0$ is a closed walk of odd length d ;
- $\chi(K^2) \leq 3$ if $d \not\equiv 2 \pmod{3}$ [2, Thm. 9], [5, Cor. 8.3];
- $\chi(K^2) \leq 4$ if $d \equiv 3 \pmod{8}$ [2, Thm. 10] or $d \equiv 0, 1, 2, 4 \pmod{7}$ [5, Cor. 8.3]; Moorhouse’s Theorem 8.1 combines these.

So a real quadratic field can need four colours only when $d \equiv 11 \pmod{12}$. In the literature we found, the largest lower bound for a real quadratic field is 3 [4, 5, 2]; explicit graphs for 26 fields that need four colours are given in the companion draft [7]. Moorhouse asked what can be said about $\chi(\mathbb{Q}(\sqrt{d})^2)$, and about $\chi(\mathbb{Q}(\sqrt{47})^2)$ in particular [6]. Fischer [2, Thm. 10(ii)] showed, with the vector u above (up to sign and mirror image), that for $d \equiv -1 \pmod{24}$ the vectors $(\frac{1}{2}, 0)$ and $(\frac{1}{3}, 0)$ are sums of unit vectors, so that the plane has no *additive* k -colouring for $k \leq 6$: none induced by a homomorphism from the group C_0 generated by the unit vectors to a group of order k (his Theorem 1 extends colourings of C_0 to the plane by cosets).

Theorem 1. $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for every positive integer $d \equiv 11 \pmod{12}$.

Corollary 2. For squarefree $d \geq 2$, $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ if and only if $d \equiv 11 \pmod{12}$. For these d , $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ unless $d \equiv 47, 143, 167 \pmod{168}$; in particular $4 \leq \chi(\mathbb{Q}(\sqrt{47})^2) \leq 5$.

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These results were obtained with the assistance of an AI system (Claude, by Anthropic), which found the construction, wrote the proofs and the programs that check them, and helped to draft this note. Separate AI agents refereed the proofs, with their own programs. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

Proof. The first statement is Theorem 1 and the bounds listed above. Among $d \equiv 11 \pmod{12}$, those not covered by Fischer's or Moorhouse's bound 4 satisfy $d \equiv 7 \pmod{8}$ and $d \equiv 3, 5, 6 \pmod{7}$, that is, $d \equiv 47, 143, 167 \pmod{168}$. For $d = 47$, the bound 5 comes from reduction at a prime above 11, which splits in $\mathbb{Q}(\sqrt{47})$, and $\chi(\mathbb{F}_{11}^2) = 5$ [5, Lemma 8.2 and Table 6.1]. $\square$

The proof is explicit. Let G_N be the set of the $4(2k+1)$ rational unit vectors whose denominators divide $N = 5^k$ (Lemma 4), and for an integer n let

$$u_n = \frac{(n + i\sqrt{d})^2}{n^2 + d} = \left(\frac{n^2 - d}{n^2 + d}, \frac{2n\sqrt{d}}{n^2 + d} \right),$$

a unit vector of $\mathbb{Q}(\sqrt{d})^2$, with mirror image $\bar{u}_n$. Rotating a vector v by $\gamma \in G_N$ gives a unit vector γv of $\mathbb{Q}(\sqrt{d})^2$.

Theorem 3. *Let $d \equiv 11 \pmod{12}$, $N = 5^k$, and let U be the following set of unit vectors.*

- (1) *If $d \equiv 23 \pmod{24}$ and $d < 21 \cdot 5^{k-1}$ (for example $5^k \geq d/4$): $U = G_N \cup G_N u_1 \cup G_N \bar{u}_1$.*
- (2) *If $d \equiv 11 \pmod{24}$, $s = v_3(d+1)$, n is a positive integer with $n \equiv 3 \pmod{4}$ and $n \equiv 1 \pmod{3^{s+1}}$ (for example $n = 1 + 2 \cdot 3^{s+1}$), and $5^k > \frac{2\sqrt{10}}{3} n(n+d)$: $U = G_N \cup G_N u_1 \cup G_N \bar{u}_1 \cup G_N u_n$.*

Then no homomorphism ξ from the group $\mathbb{Z}U$ generated by U to $\mathbb{R}/\mathbb{Z}$ maps every vector of U into the arc $[1/3, 2/3]$.

Proof of Theorem 1. By [9, Theorem 1], a Cayley graph $\text{Cay}(\Gamma, S)$ of an abelian group Γ , with $S = -S$, is 3-colourable if and only if some character $\xi: \Gamma \rightarrow \mathbb{R}/\mathbb{Z}$ maps S into $[1/3, 2/3]$. By Theorem 3, $\text{Cay}(\mathbb{Z}U, \pm U)$ is not 3-colourable. It is a subgraph of the plane over $\mathbb{Q}(\sqrt{d})$. (If d is not squarefree, its squarefree part is again $\equiv 11 \pmod{12}$, and the field is the same.) $\square$

By the theorem of de Bruijn and Erdős, each of these planes contains a finite unit-distance graph with no proper 3-colouring, and deleting vertices one at a time gives one with chromatic number exactly 4; it is triangle-free, since $\sqrt{3} \notin \mathbb{Q}(\sqrt{d})$. We do not exhibit it: the direction “3-colourable $\Rightarrow$ character” of [9, Theorem 1] averages over the group with an invariant mean and is not constructive. For 26 of these fields, explicit vertex-critical graphs with 71 to 1404 vertices and no proper 3-colouring are given in [7], with proofs that do not use characters.

Section 2 turns characters into complex numbers, Section 3 describes the set S_N of characters of $\frac{1}{N}\mathbb{Z}[i]$ that are allowed on G_N , and Sections 4 and 5 prove the two cases of Theorem 3. The bound on k is generous: for $d = 11$ the theorem needs $5^k = 3125$, but $5^k = 25$ (40 vectors up to sign) already works, as an exact certificate checked by computer shows (Section 6).

2. CHARACTERS AS COMPLEX NUMBERS

Identify $\mathbb{Q}^2$ with $\mathbb{Q}(i)$, and write a vector of $\mathbb{Q}(\sqrt{d})^2$ as $X + \sqrt{d}Y$ with $X, Y \in \mathbb{Q}(i)$: the vector $(X_1 + \sqrt{d}Y_1, X_2 + \sqrt{d}Y_2)$ corresponds to $X = X_1 + iX_2, Y = Y_1 + iY_2$. For $\gamma \in \mathbb{Q}(i)$ with $|\gamma| = 1$, multiplying X and Y by γ is the rotation by γ , an isometry of $\mathbb{Q}(\sqrt{d})^2$. In this notation $u_n = X + \sqrt{d}Y$ with $X = (n^2 - d)/(n^2 + d)$ and $Y = 2ni/(n^2 + d)$.

Let U be one of the sets of Theorem 3 and T the least common multiple of the denominators $n^2 + d$ of the vectors u_n used. Every vector of U lies in $\Gamma' = M + \sqrt{d}M$, $M = \frac{1}{NT}\mathbb{Z}[i]$, because $N\gamma \in \mathbb{Z}[i]$ for $\gamma \in G_N$ and $u_n = X + \sqrt{d}Y$ with $X, Y \in \frac{1}{T}\mathbb{Z}[i]$. Every character ξ of $\mathbb{Z}U$ extends to Γ' , since $\mathbb{R}/\mathbb{Z}$ is divisible. The characters of $\Gamma' \cong \mathbb{Z}^4$ are $X + \sqrt{d}Y \mapsto \text{Re}(\bar{c}_1 X) + \text{Re}(\bar{c}_2 Y) \pmod{1}$ with $c_1, c_2 \in \mathbb{C}$ (on the basis $\frac{1}{NT}, \frac{i}{NT}$ of M these take the arbitrary real values $\frac{\text{Re } c}{NT}, \frac{\text{Im } c}{NT}$). For a vector $v = X + \sqrt{d}Y$ put $w_v = c_1 \bar{X} + c_2 \bar{Y} \in \mathbb{C}$. Then

$$(1) \quad \xi(\gamma v) = \text{Re}(\bar{w}_v \gamma) \pmod{1} \quad (\gamma \in G_N),$$

and $v \mapsto w_v$ is $\mathbb{Q}$ -linear, so every linear relation with rational coefficients among vectors holds, exactly in $\mathbb{C}$, among the corresponding numbers w_v . Let $I = [1/3, 2/3]$ and

$$S_N = \{c \in \mathbb{C} : \text{Re}(\bar{c}\gamma) \in I + \mathbb{Z} \text{ for every } \gamma \in G_N\}.$$

If $\xi(U) \subseteq I + \mathbb{Z}$, then $w_v \in S_N$ for each $v \in \{1, u_1, \bar{u}_1, u_n\}$, where $1 = (1, 0)$. The set S_N is invariant under $N\mathbb{Z}[i]$ (as $N\gamma \in \mathbb{Z}[i]$), under $c \mapsto -c$ and under multiplication by i .

3. THE SHAPE OF S_N

The definitions of G_N and S_N make sense for every $k \geq 0$; $G_1 = \{\pm 1, \pm i\}$. Let $\rho = (3+4i)/5 = (2+i)/(2-i)$, $c_N^* = N(1+i)/2$, $Q_N = \{\frac{N}{3}(\alpha + \beta i) : \alpha, \beta \in \{1, 2\}\}$, $S_{q_0} = \{x + iy : |x|, |y| \leq 1/6\}$, and for $k \geq 0$

$$P_k = \{x \in \mathbb{C} : \bar{x}\rho^j \in S_{q_0} \text{ for all integers } |j| \leq k\},$$

a closed convex polygon; $P_0 = S_{q_0}$ and $P_k \subseteq P_1$ for $k \geq 1$. Let $\Lambda_{\pm} = \frac{2 \pm i}{5}\mathbb{Z}[i]$; both contain $\mathbb{Z}[i]$ with index 5.

Lemma 4. $G_N = \{\varepsilon\rho^j : \varepsilon \in \{\pm 1, \pm i\}, |j| \leq k\}$, a set of $4(2k+1)$ elements.

Proof. Let $\gamma \in G_N$, $\gamma = \alpha/5^t$ with $\alpha \in \mathbb{Z}[i]$ and $t \leq k$ least. Then $|\alpha|^2 = 25^t$, so $\alpha = \varepsilon(2+i)^e(2-i)^f$ with $e+f = 2t$; if $t \geq 1$, minimality excludes $e, f \geq 1$, so $\gamma = \varepsilon(2 \pm i)^{2t}/5^t = \varepsilon\rho^{\pm t}$. These elements are distinct, since ρ^j has exact denominator $5^{|j|}$. $\square$

Lemma 5. (a) $\bar{c}_N^*\gamma \in \frac{1+i}{2} + \mathbb{Z}[i]$ for every $\gamma \in G_N$. Hence $c_N^* + x \in S_N$ if and only if $\bar{x}\gamma \in S_{q_0} + \mathbb{Z}[i]$ for every $\gamma \in G_N$; in particular $c_N^* + P_k \subseteq S_N$. (b) $Q_N \subseteq S_N$.

Proof. (a) By Lemma 4, $\bar{c}_N^*\gamma = \frac{5^{k-|j|}}{2}(1-i)\varepsilon\pi$ with $\pi = (2 \pm i)^{2|j|}$. As $\varepsilon\pi$ has odd norm, $\varepsilon\pi \equiv 1 \pmod{1+i}$ and $(1-i)\varepsilon\pi \equiv 1-i \pmod{2}$, so both coordinates are odd. Since $I = \frac{1}{2} + [-\frac{1}{6}, \frac{1}{6}]$, the conditions for γ and $i\gamma$ together say $\bar{x}\gamma \in S_{q_0} + \mathbb{Z}[i]$. (b) For $q = \frac{N}{3}(\alpha + \beta i)$, $\bar{q}\gamma = \frac{5^{k-|j|}}{3}(\alpha - \beta i)\varepsilon\pi$. In $\mathbb{Z}[i]/3 = \mathbb{F}_9$, the elements with both coordinates nonzero, $\pm 1 \pm i$, are the non-squares, since $1+i$ generates $\mathbb{F}_9^\times$ and $(1+i)\mu_4 = \{\pm 1 \pm i\}$. Now $\alpha - \beta i$ is a non-square and $\varepsilon\pi$ a nonzero square, so both coordinates of $(\alpha - \beta i)\varepsilon\pi$ are prime to 3, and those of $\bar{q}\gamma$ lie in $\{\frac{1}{3}, \frac{2}{3}\} + \mathbb{Z}$. $\square$

Lemma 6. (a) $\max_{x \in P_1} |x| = \sqrt{10}/18 \approx 0.1757$. (b) Every point of $(\Lambda_+ \cup \Lambda_-) \setminus \mathbb{Z}[i]$ is at distance at least $\sqrt{2}/6 \approx 0.2357$ from $S_{q_0} + \mathbb{Z}[i]$, and none lies in $[-\frac{1}{3}, \frac{1}{3}]^2 + \mathbb{Z}[i]$.

Proof. (a) P_1 is cut out by the six slabs $|\langle x, \nu \rangle| \leq \frac{1}{6}$, $\nu \in \{1, i, \rho, i\rho, \bar{\rho}, i\bar{\rho}\}$, whose directions modulo 180° are $0^\circ, \theta, 90^\circ - \theta, 90^\circ, 90^\circ + \theta, 180^\circ - \theta$ with $\theta = \arccos \frac{4}{5}$. So P_1 is a 12-gon circumscribed about the circle of radius $\frac{1}{6}$; consecutive normals make angles θ (four times) and $90^\circ - 2\theta$ (twice), and the vertices are at distance $\frac{1}{6}/\cos \frac{\theta}{2} = \frac{1}{6} \cdot \frac{\sqrt{10}}{3}$, for example $(\frac{1}{6}, \frac{1}{18})$, or closer. (b) The nonzero classes of $\Lambda_+/\mathbb{Z}[i]$ are those of $\pm \frac{2+i}{5}$ and $\pm \frac{-1+2i}{5}$, whose representatives in $(-\frac{1}{2}, \frac{1}{2})^2$ have one coordinate $\pm \frac{2}{5}$ and the other $\pm \frac{1}{5}$. The point of S_{q_0} nearest to $\frac{2}{5} + \frac{i}{5}$ is $\frac{1}{6} + \frac{i}{6}$, at distance $\sqrt{(7/30)^2 + (1/30)^2} = \sqrt{2}/6$; other translates of S_{q_0} are farther. The coordinate $\pm \frac{2}{5}$ is at distance $\frac{2}{5} > \frac{1}{3}$ from $\mathbb{Z}$. Λ_- is the mirror image of Λ_+ . $\square$

Lemma 7 (unique lifts). Let $k \geq 1$, $N = 5^k$, $N' = 5^{k-1}$. If $c, c' \in S_N$ and $c' - c \in N'\mathbb{Z}[i]$, then $c' - c \in N\mathbb{Z}[i]$.

Proof. Write $c' = c + N'\lambda$. Then $\bar{c}'\rho^k - \bar{c}\rho^k = N'\bar{\lambda}\rho^k = \bar{\lambda}(2+i)^{2k-1}/(2-i) \in \Lambda_+$. Both $\bar{c}\rho^k$ and $\bar{c}'\rho^k$ lie in $(I + \mathbb{Z}) + i(I + \mathbb{Z})$ (the conditions for ρ^k and $i\rho^k$), so the difference lies in $[-\frac{1}{3}, \frac{1}{3}]^2 + \mathbb{Z}[i]$, hence in $\mathbb{Z}[i]$ by Lemma 6(b): that is, $(2-i) \mid \bar{\lambda}$, or $(2+i) \mid \lambda$. With ρ^{-k} and Λ_- , $(2-i) \mid \lambda$. So $5 \mid \lambda$. $\square$

Proposition 8. For every $k \geq 1$, $S_{5^k} = (c_N^* + P_k + N\mathbb{Z}[i]) \cup (Q_N + N\mathbb{Z}[i])$. So every $c \in S_N$ is $c = e + x$ with either $e \in c_N^* + N\mathbb{Z}[i]$ and $|x| \leq \sqrt{10}/18$, or $e \in Q_N + N\mathbb{Z}[i]$ and $x = 0$.

Proof. “ $\supseteq$ ” is Lemma 5. For “ $\subseteq$ ” we induct on k ; write $N' = 5^{k-1}$ and note that $S_N \subseteq S_{N'}$ and $c_N^* - c_{N'}^* = 2N'(1+i) \in N'\mathbb{Z}[i]$.

$k = 1$. $S_1 = \frac{1+i}{2} + S_{q_0} + \mathbb{Z}[i]$, so $c \in S_5$ is $c = c_5^* + x + \lambda$ with $x \in S_{q_0}$, $\lambda \in \mathbb{Z}[i]$. By Lemma 5(a), the conditions at ρ say $\bar{x}\rho + \bar{\lambda}\rho \in S_{q_0} + \mathbb{Z}[i]$, where $\bar{\lambda}\rho \in \Lambda_+$. If $\bar{\lambda}\rho \notin \mathbb{Z}[i]$, Lemma 6(b) gives $|x| = |\bar{x}\rho| \geq \sqrt{2}/6$, and as $x \in S_{q_0}$, x is a corner $(\pm 1 \pm i)/6$; likewise with $\bar{\rho}$. If x is not a corner, then $(2+i) \mid \lambda$ and $(2-i) \mid \lambda$, so $c \equiv c_5^* + x \pmod{5\mathbb{Z}[i]}$, and $\bar{x}\rho^{\pm 1} \in S_{q_0} + \mathbb{Z}[i]$ with $|x| < \frac{5}{6}$

gives $\bar{x}\rho^{\pm 1} \in \text{Sq}_0$: $x \in P_1$. If x is a corner, $c \equiv \frac{1+i}{2} + \frac{\pm 1 \pm i}{6} = \frac{\alpha + \beta i}{3} \pmod{\mathbb{Z}[i]}$ with $\alpha, \beta \in \{1, 2\}$; the point $q = \frac{5}{3}(\alpha' + \beta' i) \in Q_5$ with $\alpha' \equiv 2\alpha, \beta' \equiv 2\beta \pmod{3}$ lies in S_5 and $q \equiv c \pmod{\mathbb{Z}[i]}$, so $c \in q + 5\mathbb{Z}[i]$ by Lemma 7.

$k \geq 2$. Let $c \in S_N \subseteq S_{N'}$. If $c \equiv \frac{N'}{3}(\alpha + \beta i) \pmod{N'\mathbb{Z}[i]}$, then $q = \frac{N}{3}(\alpha' + \beta' i) \in Q_N$ with $\alpha' \equiv 2\alpha, \beta' \equiv 2\beta \pmod{3}$ satisfies $q \equiv c \pmod{N'\mathbb{Z}[i]}$ and $q \in S_N$, so $c \in q + N\mathbb{Z}[i]$ by Lemma 7. Otherwise $c = c_N^* + x + N'\lambda$ with $x \in P_{k-1} \subseteq P_1$ and $\lambda \in \mathbb{Z}[i]$, and the conditions at ρ^k say $\bar{x}\rho^k + \mu \in \text{Sq}_0 + \mathbb{Z}[i]$ with $\mu = N'\bar{\lambda}\rho^k \in \Lambda_+$. If $\mu \notin \mathbb{Z}[i]$, Lemma 6 gives $\sqrt{2}/6 \leq |x| \leq \sqrt{10}/18$, which is impossible. So $(2+i) \mid \lambda$, likewise $(2-i) \mid \lambda$, hence $c \equiv c_N^* + x \pmod{N\mathbb{Z}[i]}$, and $\bar{x}\rho^{\pm k} \in \text{Sq}_0$, so $x \in P_k$. $\square$

Types. By Proposition 8 write each $c \in S_N$ as $c = N\varepsilon + x$ with $|x| \leq r := \sqrt{10}/18$ and ε of type c , $\varepsilon \in \frac{1+i}{2} + \mathbb{Z}[i]$, or of type q , $\varepsilon \in \frac{\alpha + \beta i}{3} + \mathbb{Z}[i]$ with $\alpha, \beta \in \{1, 2\}$; in both cases $N\varepsilon \in \frac{N}{6}\mathbb{Z}[i]$. Let $\mathbb{Z}_{(p)}[i]$ be the elements of $\mathbb{Q}(i)$ whose denominators are prime to p , and v_3 the 3-adic valuation of $\mathbb{Q}(i)$ (3 is inert in $\mathbb{Z}[i]$). Then: ε has type q if and only if $v_3(\varepsilon) = -1$; type c if and only if $\varepsilon \notin \mathbb{Z}_{(2)}[i]$, and then $\varepsilon \equiv h := \frac{1+i}{2} \pmod{\mathbb{Z}_{(2)}[i]}$; $mh \equiv h$ for odd m , and $2h \in \mathbb{Z}[i]$.

Lemma 9 (exact relations). *Let $\sigma_1, \dots, \sigma_\ell \in S_N$, $\sigma_j = N\varepsilon_j + x_j$, and let $m_1, \dots, m_\ell \in \mathbb{Z}$ with $\sum m_j \sigma_j = 0$. If $N > 6r \sum |m_j|$, then $\sum m_j \varepsilon_j = 0$.*

Proof. $\sum m_j N\varepsilon_j = -\sum m_j x_j$; the left side lies in $\frac{N}{6}\mathbb{Z}[i]$, whose nonzero elements have modulus at least $\frac{N}{6}$, and the right side has modulus at most $r \sum |m_j| < \frac{N}{6}$. $\square$

4. THE CASE $d \equiv 23 \pmod{24}$

Let $a = \frac{d+1}{4}$ and $b = \frac{d-1}{2}$: here $a \equiv 0 \pmod{6}$, b is odd and $3 \nmid b$. The relation $a(u_1 + \bar{u}_1) + b \cdot 1 = 0$ gives, for $z = w_1$, $w = w_{u_1}$, $w' = w_{\bar{u}_1}$ in S_N ,

$$a(w + w') + bz = 0.$$

By Proposition 8 write $w = e_1 + x_1$, $w' = e_2 + x_2$, $z = e_3 + y$, with $e_j \in (c_N^* + N\mathbb{Z}[i]) \cup (Q_N + N\mathbb{Z}[i])$, $x_1, x_2, y \in P_k$, and the corresponding x_j or y equal to 0 when $e_j \in Q_N + N\mathbb{Z}[i]$. Then

$$a(e_1 + e_2) + be_3 = -(ax_1 + ax_2 + by).$$

As a is even and $3 \mid a$, $ac_N^* = \frac{a}{2}N(1+i)$ and $a\frac{N}{3}(\alpha + \beta i)$ lie in $N\mathbb{Z}[i]$; as b is odd and $3 \nmid b$, be_3 lies in $c_N^* + N\mathbb{Z}[i]$ or in $Q_N + N\mathbb{Z}[i]$ with e_3 . So the left side $u + iv$ lies in $c_N^* + N\mathbb{Z}[i]$, where $|u|, |v| \geq \frac{N}{2}$, or in $Q_N + N\mathbb{Z}[i]$ ($-Q_N \equiv Q_N$), where $|u|, |v| \geq \frac{N}{3}$. As P_k is convex and symmetric, the right side lies in $(2a+b)P_k = dP_k$ in the first case and in $2aP_k = \frac{d+1}{2}P_k$ in the second ($y = 0$). On $P_1 \supseteq P_k$ the conditions at ρ and $\bar{\rho}$ give $|3u \pm 4v| \leq \frac{5}{6}$, so $3|u| + 4|v| \leq \frac{5}{6}$. Hence the first case needs $\frac{7N}{2} \leq \frac{5d}{6}$, that is $d \geq \frac{21N}{5}$, and the second needs $\frac{7N}{3} \leq \frac{5(d+1)}{12}$, that is $d+1 \geq \frac{28N}{5}$. If $d < 21N/5$, neither holds. This proves Theorem 3(1); and $5^k \geq d/4$ gives $d \leq 4N < 21N/5$.

Remark. The bound is sharp. An exact computation (cells of S_N , Minkowski sums and the lattice points of $N\mathbb{Z}[i]$, with a second method as a cross-check and an explicit character in every feasible case) shows, for every $d \equiv 23 \pmod{24}$ up to 1295 when $N = 5, 25, 125$ and up to 2999 when $N = 625$, that the set of Theorem 3(1) admits no such character exactly when $d < 21N/5$.

5. THE CASE $d \equiv 11 \pmod{24}$

Now a is odd, and the argument fails: $a+b$ is even and $3 \mid a$, so $a(c_N^* + q) + bc_N^* \in N\mathbb{Z}[i]$ for $q \in Q_N$, and since $\gcd(a, b) = 1$ suitable lifts $z = c_N^* + N\lambda_3$, $w = c_N^* + N\lambda_1$, $w' = q + N\lambda_2$ solve $a(w + w') + bz = 0$ exactly. So $G_N \cup G_N u_1 \cup G_N \bar{u}_1$ admits such a character for every k . The vector u_n excludes it. Comparing X and Y gives $(n^2 + d)u_n = (n-1)(n+d) \cdot 1 + n(1+d)u_1$, so with $y = w_{u_n}$,

$$a(w + w') + bz = 0, \quad (n^2 + d)y - (n-1)(n+d)z - n(1+d)w = 0.$$

As $n \geq 1$, all coefficients are nonnegative, and the sums of their absolute values are $2a + b = d$ and $2n(n+d) \geq d$. (For negative n the second sum is larger, and the hypothesis on N would have

to change.) As $N > \frac{2\sqrt{10}}{3}n(n+d) = 12rn(n+d)$, Lemma 9 applies to both relations: writing $z = N\varepsilon_z + x_z$ and so on,

$$(2) \quad a(\varepsilon_w + \varepsilon_{w'}) = -b\varepsilon_z, \quad (n^2 + d)\varepsilon_y = (n-1)(n+d)\varepsilon_z + n(1+d)\varepsilon_w,$$

exactly in $\mathbb{Q}(i)$.

Step 1: ε_z has type c and $v_3(\varepsilon_z) \geq s-1$. We have $v_3(a) = s \geq 1$, $3 \nmid b$, and $v_3(\varepsilon_w + \varepsilon_{w'}) \geq -1$, so $v_3(\varepsilon_z) \geq s-1 \geq 0$.

Step 2 (at 2): ε_y has type c if and only if ε_w has type q . As $d \equiv 3 \pmod{8}$ and $n \equiv 3 \pmod{4}$, each of $n^2 + d$, $(n-1)(n+d)$ and $n(1+d)$ is 4 times an odd integer; write the second relation as $A'\varepsilon_y = B'\varepsilon_z + C'\varepsilon_w$ with A', B', C' odd. Modulo $\mathbb{Z}_{(2)}[i]$, $B'\varepsilon_z \equiv h$, and $C'\varepsilon_w \equiv h$ or 0 according as ε_w has type c or q . So $A'\varepsilon_y \in \mathbb{Z}_{(2)}[i]$ exactly when ε_w has type c .

Step 3 (at 3): ε_y has type q if and only if ε_w has type q . Let $e = v_3(n-1) \geq s+1$; $3 \nmid n(n+1)$. Then $n^2 + d = (n-1)(n+1) + (d+1)$ and $n+d = (n-1) + (d+1)$ have valuation s , so do $n(1+d)$, and $v_3((n-1)(n+d)) = e + s$. Dividing the second relation by 3^s gives $A''\varepsilon_y = B''\varepsilon_z + C''\varepsilon_w$ with $3 \nmid A''C''$ and $v_3(B''\varepsilon_z) \geq e + s - 1 \geq 2s \geq 2$. So $v_3(\varepsilon_y) = -1$ if and only if $v_3(\varepsilon_w) = -1$.

Steps 2 and 3 contradict each other whatever the type of ε_w . This proves Theorem 3(2).

Remark. One vector cannot replace u_1 and u_n : if $v = (\mu + i\nu\sqrt{d})/\tau$ with $\mu^2 + d\nu^2 = \tau^2$ in lowest terms and $d \equiv 3 \pmod{8}$, then $4 \nmid \tau$ (if τ is even, μ, ν are odd and $\mu^2 + d\nu^2 \equiv 4 \pmod{8}$), while the argument of Section 4 needs the coefficient $\tau/\gcd(\tau, 2\mu)$ of the relation $\tau(v + \bar{v}) = 2\mu \cdot 1$ to be even. Correspondingly, $x^2 + y^2$ is anisotropic over $\mathbb{Q}_2(\sqrt{d})$ for $d \equiv 3 \pmod{8}$ and isotropic for $d \equiv 7 \pmod{8}$.

6. CHECKS

Every computation that the proofs use is checked in exact arithmetic by a program in [8]: Lemmas 4 and 5 for $k \leq 6$, both numbers of Lemma 6, and S_{5^k} itself for $k = 1, 2, 3$ by exact polygon clipping (a 12-, 20- and 28-gon around c_N^* of areas 17/189, 61/693 and 437/4977, and the four points of Q_N), independently of the induction; a referee's own program confirmed Proposition 8 with equality for $k \leq 4$, and found no exception among 10^7 sampled points for $k = 1, 2$. Exact certificates (branch and bound over the integer values of a basis of the relations, with a rational Farkas vector at every leaf), each checked by two independent checkers, show that the sets of Theorem 3 with smaller N than the proof needs already have no such character: in case (1), $d = 23, 47, 71, 95$ with $N = 25$ and $d = 119, 143, \dots, 311$ ($d \equiv 23 \pmod{24}$, squarefree) with $N = 125$; in case (2), $d = 11, 35, 59$ with $N = 25$ and $d = 83, 131, 155$ with $N = 125$. A floating-point solver confirms case (2) with the N that the proof needs for $d = 11, 35, 59, 83, 131, 179$. Finally, for a finite set Δ of unit vectors, Proposition 8 and Lemma 9 reduce the question whether $G_N(\Delta \cup \{1\})$ admits such a character for all large N to the existence of a character with values in $\{\frac{1}{3}, \frac{1}{2}, \frac{2}{3}\}$, a computation in $(\mathbb{Z}/6)^4$; it confirms Theorem 3(2) for all 191 squarefree $d \equiv 11 \pmod{24}$ below 5000, and shows that for $d = 11$ no two vectors $(\mu + i\nu\sqrt{11})/\tau$ with $\nu \leq 7$, $\tau \leq 60$ suffice.

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